

Final Project Report Template

1. Introduction

The Food Ordering Data Visualization Dashboard is a Tableau project that converts raw food ordering data into clear and interactive visualizations. It helps analyze important aspects such as customer preferences, sales performance, restaurant performance, delivery efficiency, and ordering trends. By presenting the data through charts and graphs, the dashboard makes it easier to identify patterns and gain useful insights, supporting better decision-making in the food delivery industry.

1.1. Project overviews

The project aims to create an interactive dashboard that provides a clear overview of food ordering data. It includes different visualizations to analyze customer ordering patterns, restaurant performance, delivery efficiency, and overall order trends. Interactive filters allow users to explore the data from different perspectives, making the dashboard useful for understanding business performance and supporting better decision-making.

1.2. Objectives

- To design an interactive and user-friendly dashboard using Tableau.
- To analyze food ordering data and identify useful business insights.
- To track key metrics such as total orders, order value, ratings, and delivery performance.
- To understand customer preferences and ordering behavior.
- To compare restaurant performance and identify important trends.
- To use interactive visualizations for easy data exploration and analysis.
- To support better business decisions through data-driven insights.
- To demonstrate how data visualization can transform raw data into meaningful information.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Restaurant Manager	Understand customer ordering patterns and identify peak demand periods.	It is difficult to analyze large volumes of order data manually.	The data is spread across multiple attributes such as city, cuisine, meal type, order time, and customer demographics.	Unable to make quick, data-driven decisions for improving restaurant operations and customer satisfaction.
PS-2	Business Analyst / Marketing Manager	Improve customer retention and increase sales through better marketing strategies.	It is challenging to identify the factors influencing repeat orders, customer spending, discounts, and ordering behavior.	There is no single interactive dashboard that combines customer behavior, restaurant performance, discounts, ratings, and weather impact.	Difficult to design effective promotional campaigns and make informed business decisions.

2.2. Project Proposal (Proposed Solution)

Project Overview	
Objective	To analyze food ordering behavior using Tableau and create interactive dashboards and stories that provide insights into customer preferences, ordering trends, and business performance.
Scope	The project includes data cleaning, analysis, visualization, dashboard creation, and story development in Tableau. It covers customer demographics, order patterns, cuisines, discounts, weather impact, and repeat orders. It does not include real-time data collection or predictive analytics. This follows the same sections as the provided proposal template.
Problem Statement	
Description	Food delivery businesses generate large amounts of order data, making it difficult to identify customer preferences, peak ordering times, popular cuisines, and factors affecting order values through raw data alone.
Impact	Analyzing this data helps businesses understand customer behavior, improve marketing strategies, optimize restaurant offerings, and make better business decisions through interactive visualizations. This corresponds to the "Description" and "Impact" fields in the template.
Proposed Solution	

Approach	Clean and prepare the dataset, analyze customer ordering patterns in Tableau, develop interactive dashboards and stories, and present insights using different visualizations such as bar charts, heat maps, pie charts, and line charts.
Key Features	• Interactive Tableau dashboards • Tableau Story with key business insights • Analysis of customer demographics and ordering patterns • Visualization of discounts, repeat orders, cuisines, cities, and weather impact

2.3. Initial Project Planning

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Sprint Start Date	Sprint End Date (Planning)
Sprint 1	Data Collection	USN-1	As a data analyst, I want to collect food ordering data so that I can analyze customer ordering behavior.	2	High	18 th July 2026	19 th July 2026
Sprint 2	Data Preparation	USN-2	As a data analyst, I want to clean and organize the dataset so that it is ready for visualization.	2	High	18 th July 2026	19 th July 2026
Sprint 3	Data Analysis	USN-3	As a data analyst, I want to identify customer ordering trends by city, age group, and time so that business insights can be generated.	3	High	18 th July 2026	19 th July 2026
Sprint 4	Dashboard Development	USN-4	As a business user, I want interactive Tableau dashboards so that I can monitor food ordering patterns easily.	3	Medium	18 th July 2026	19 th July 2026

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Section	Description
Project Overview	The Food Ordering Behavior Analysis project aims to analyze customer food ordering data using Tableau. The project focuses on identifying ordering patterns, customer preferences, city-wise performance, cuisine popularity, restaurant performance, discount impact, repeat orders, and weather influence. Interactive dashboards and Tableau Stories are created to transform raw data into meaningful business insights. This follows the "Project Overview" section of the template.
Data Collection Plan	The food ordering dataset was collected in CSV format from a publicly available dataset source. It contains customer demographics, order details, restaurant information, cuisine types, ratings, discounts, weather conditions, and repeat order information. The dataset was imported into Tableau, where it was cleaned, validated, and prepared for visualization and analysis. This corresponds to the "Data Collection Plan" section in the template.
Raw Data Sources Identified	Two main data environments have been identified to support the analytical stack:

	1. Behavioral Risk Profiles Database: Captured clinical factors regarding personal habits and body mass metrics across patient categories. 2. Institutional Clinical Records: Localized demographic registries tracking regional health departments, tracking metrics like clinical attendance, group tracking, and structural logs.
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3.2. Data Quality Report

Data Source	Data Quality Issue	Severity	Resolution Plan
Data Quality Issue	Inconsistent Text Formatting: Some categorical fields such as City, Cuisine, Restaurant Type, and Meal Type contained inconsistent capitalization, extra spaces, and formatting variations	Moderate	Applied text cleaning by removing leading/trailing spaces, correcting capitalization, and standardizing category names to ensure accurate grouping and visualization in Tableau.

3.3. Data Exploration and Preprocessing

Section	Description
Data Overview	The dataset contains customer food ordering records, including customer demographics, city, cuisine, meal type, restaurant type, order value, delivery fee, ratings, discounts, weather conditions, and repeat order information. It is used to analyze customer ordering behavior and business trends. This corresponds to the "Data Overview" section in the template.
Data Cleaning	Missing Values: Checked for missing values and removed or corrected incomplete records where necessary. Duplicates: Verified that duplicate order records were not present to maintain data accuracy. Error Correction: Corrected inconsistent text formatting, removed extra spaces, and standardized category names for reliable analysis. This follows the template's "Data Cleaning" section.
Data Transformation	Calculated Fields: Created calculated measures such as average order value, count of orders, and percentage of repeat orders. Aggregations: Used COUNT, SUM, and AVG functions in Tableau to summarize and compare customer ordering patterns. Filtering: Applied filters for city, age group, cuisine, meal type, weather, and discount status to support interactive analysis. This aligns with the "Data Transformation" section.
Data Type Conversion	Verified that Order ID and User ID were treated as numeric identifiers, Order Value and Rating as numeric measures, and categorical fields such as City, Cuisine, Meal Type, Restaurant

	Type, and Discount Applied as dimensions for visualization. This matches the "Data Type Conversion" section
Column Splitting and Merging	No column splitting or merging was required, as the dataset was already organized with separate fields for each attribute and ready for analysis in Tableau. This corresponds to the template's "Column Splitting and Merging" section.
Data Modeling	The cleaned dataset was imported into Tableau Public, where dimensions and measures were logically organized to create dashboards and stories for analyzing customer behavior, ordering trends, and business insights. This corresponds to the "Data Modeling" section.
Save Processed Data	The processed dataset was saved in CSV format and imported into Tableau Public for visualization. The Tableau workbook containing dashboards and stories was saved for future updates and presentation. This follows the "Save Processed Data" section in the template.

4. Data Visualization

4.1. Framing Business Questions

Q1: Which cities are performing best in terms of orders?

- Visualization: Horizontal Bar Chart
- Insight Provided: Compares the total number of food orders across different cities to identify the highest-performing locations. Mumbai records the highest number of orders, followed by Bangalore and Delhi.

Q2: When are customers ordering food the most?

- Visualization: Heat Map

- Insight Provided: Displays customer ordering patterns across weekdays, weekends, and different times of the day. Weekend evenings and nights show the highest order activity.

Q3: Which cuisines are preferred during different meals?

- Visualization: Highlight Table
- Insight Provided: Compares cuisine preferences across breakfast, lunch, dinner, and snacks. Customer preferences remain fairly balanced across all meal types.

Q4: Which age group contributes the most orders?

- Visualization: Horizontal Bar Chart
- Insight Provided: Shows that customers aged 26–35 place the highest number of food orders, making them the primary customer segment.

Q5: How does customer mood influence average order value and ratings?

- Visualization: Scatter Plot
- Insight Provided: Compares average order value and average customer ratings across different customer moods, showing only slight variations.

Q6: Do discounts affect the average order value?

- Visualization: Vertical Bar Chart
- Insight Provided: Compares average order values for discounted and non-discounted orders, indicating that discounts have minimal impact on customer spending.

Q7: What percentage of orders are repeat orders?

- Visualization: Pie Chart
- Insight Provided: Displays the proportion of repeat and non-repeat orders, showing an almost equal distribution between the two categories.

Q8: Which restaurant type generates the highest total order value?

- Visualization: Bar Chart

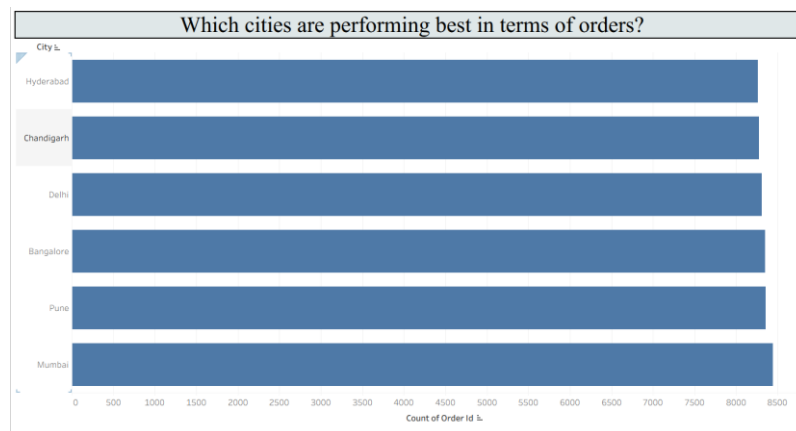
- Insight Provided: Compares total order values across Budget, Mid-range, and Premium restaurants. Premium restaurants generate the highest total order value.

Q9: Does rainy weather influence average order value?

- Visualization: Line Chart
- Insight Provided: Compares average order values during rainy and non-rainy weather across different times of the day. Rainy weather has little impact on customer spending.

4.2. Developing Visualizations

Ques1.



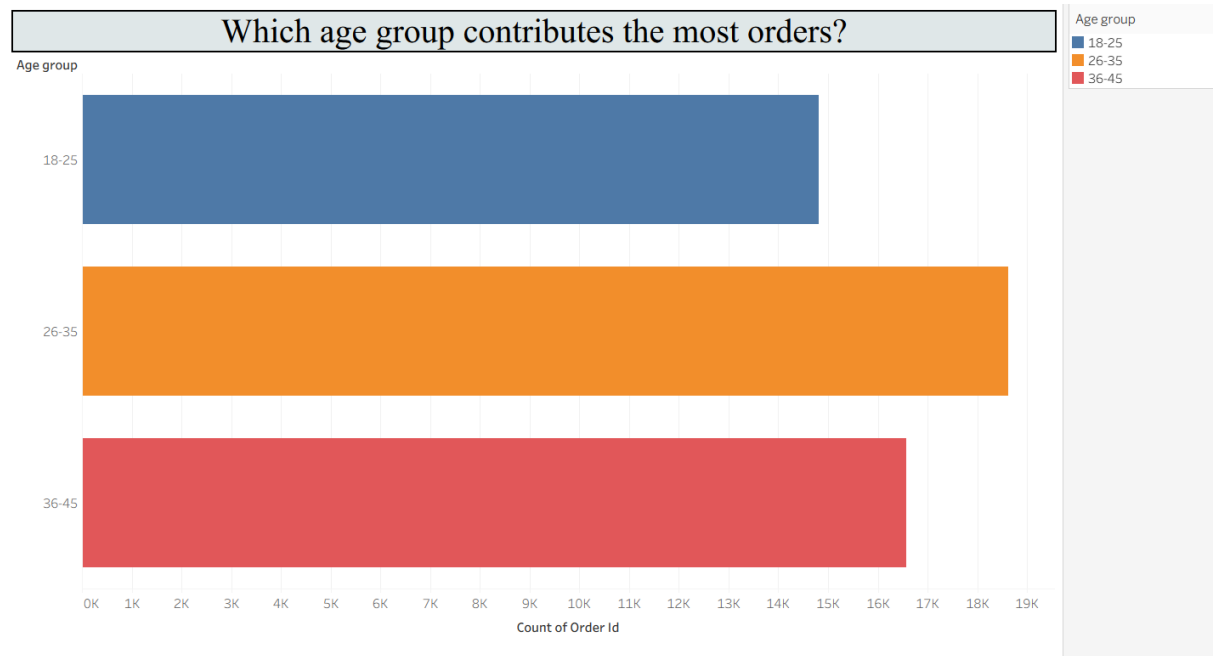
Ques 2.



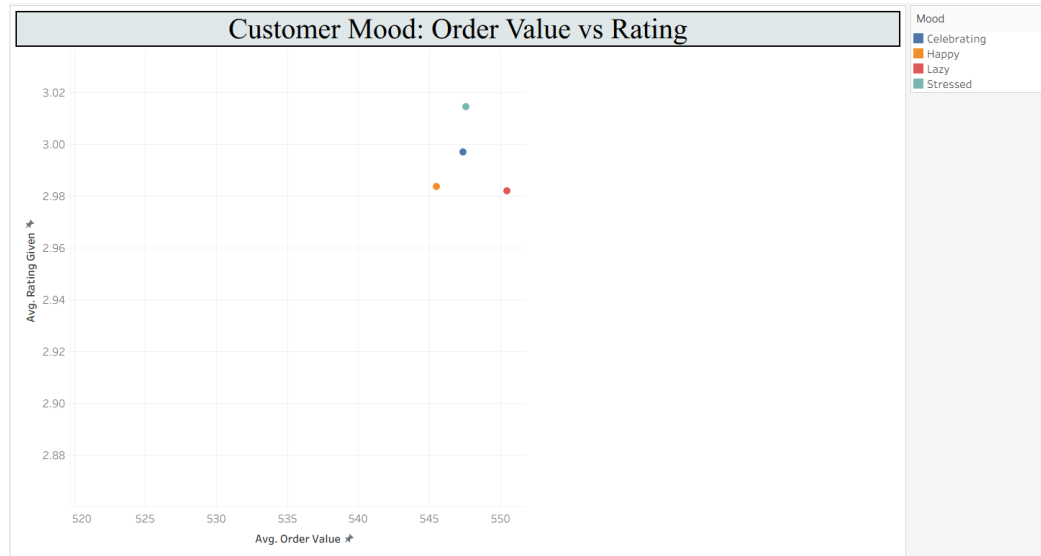
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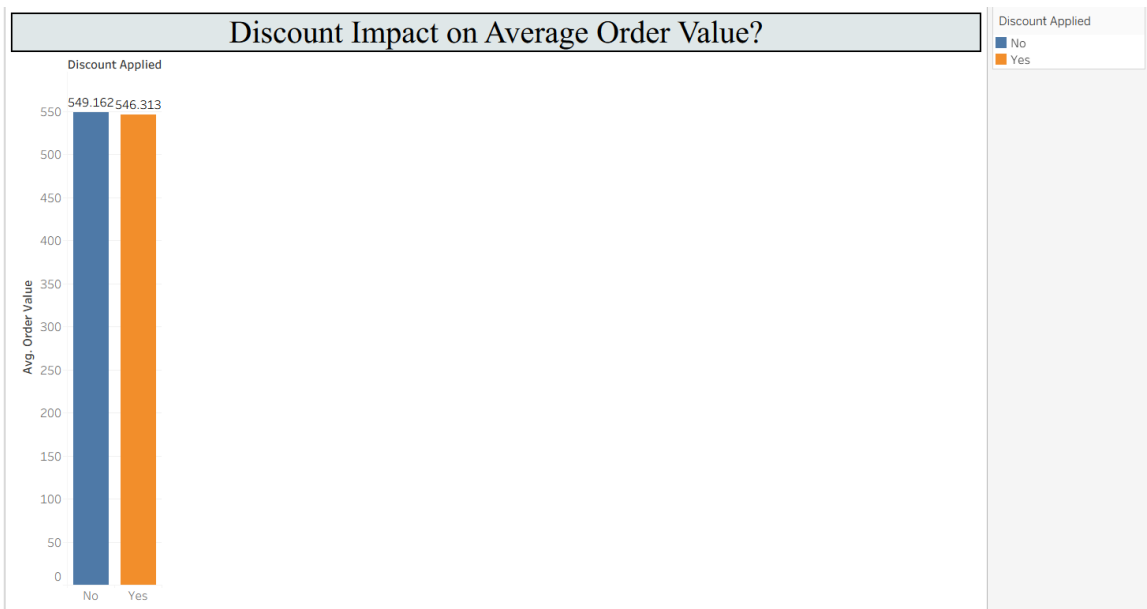
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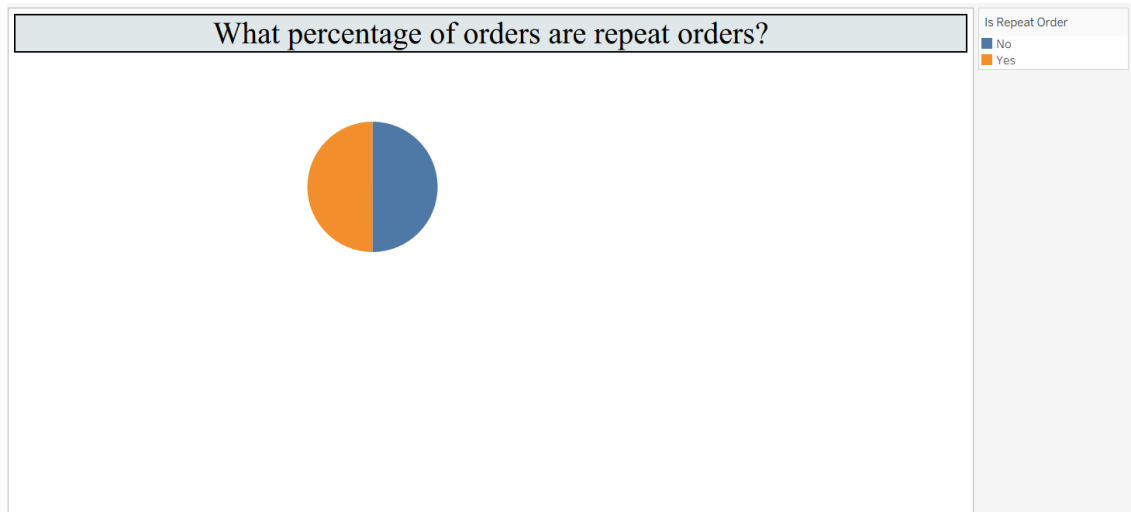
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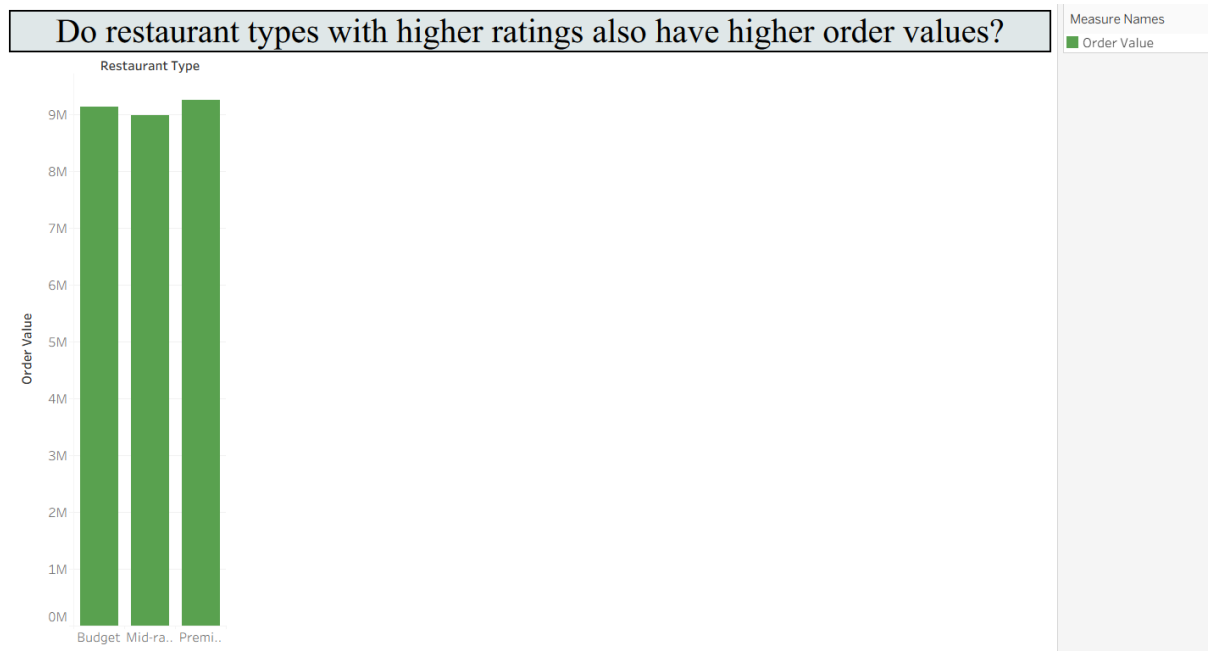
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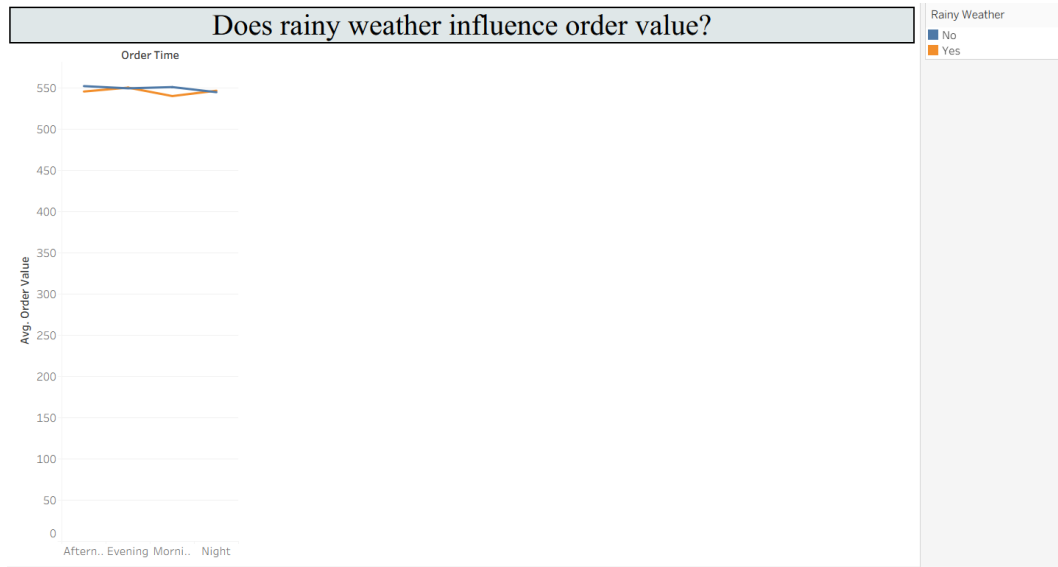
Ques 7.



Ques 8.



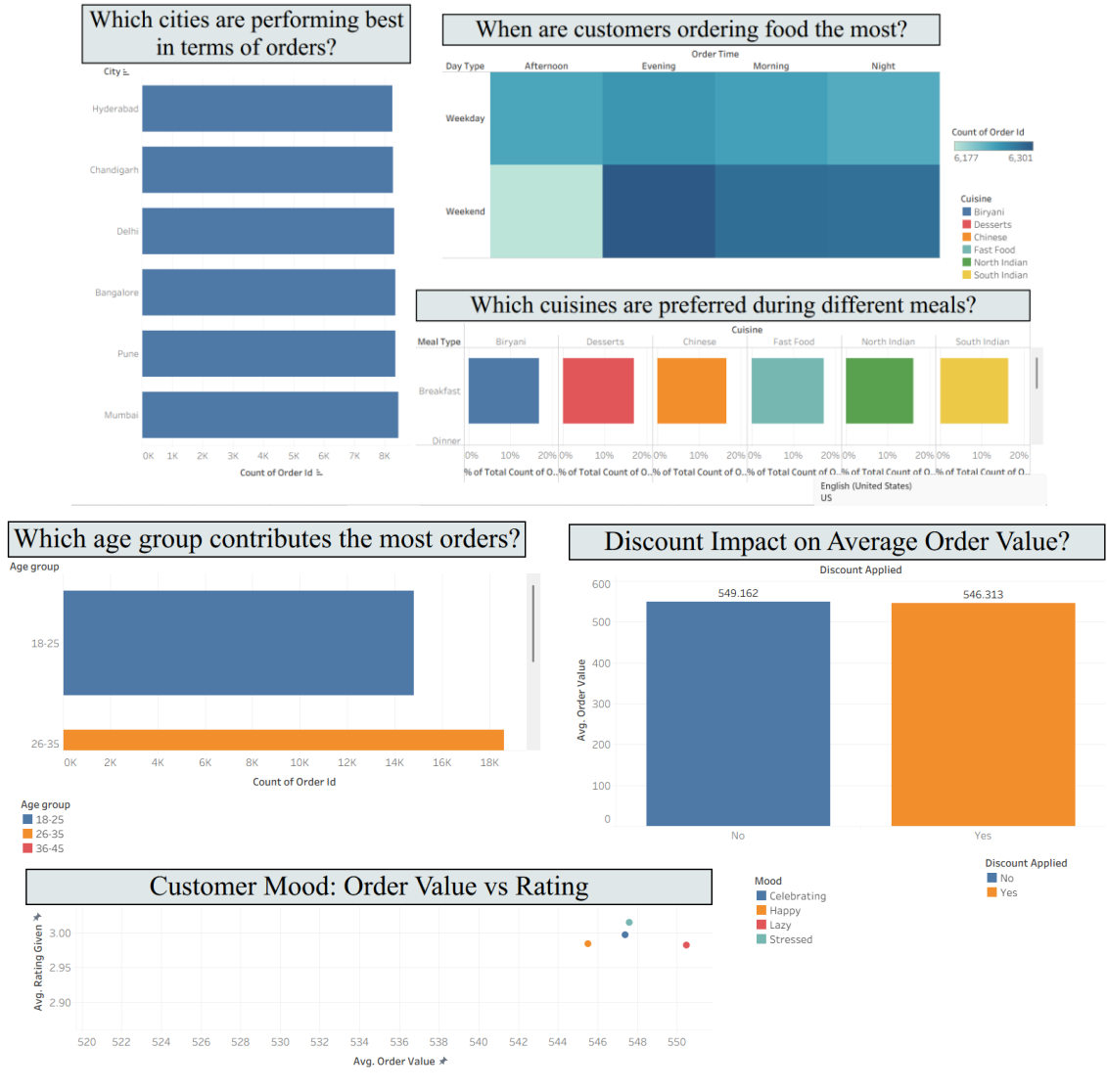
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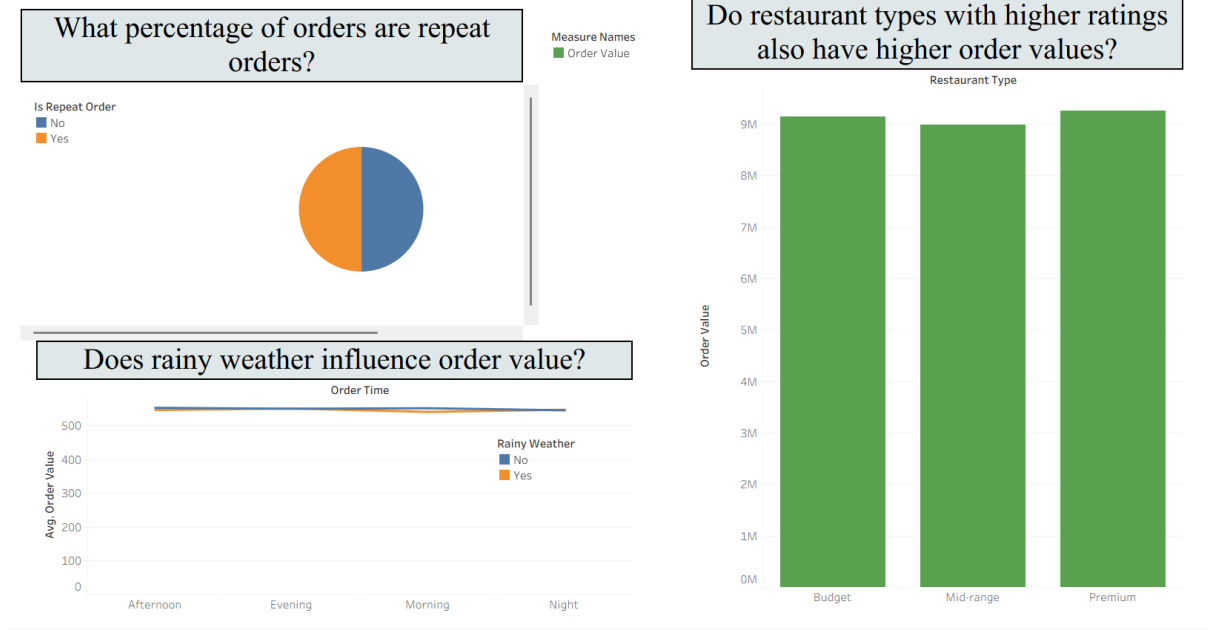


5. Dashboard

5.1. Dashboard Design File

Activity 1: Interactive and Visually Appealing Dashboards





The dashboard was designed using Tableau by applying key visualization principles such as **clear layout**, **consistent color themes**, **interactive filters**, and **user-friendly navigation**. The dashboards provide meaningful insights into customer ordering behavior and support business decision-making through interactive visualizations. This aligns with the dashboard design activity described in the template.

Dashboard 1 – Customer Order Trends

- **City-wise Order Analysis:** The horizontal bar chart compares the total number of orders across different cities, helping identify the highest-performing locations.
- **Order Timing Analysis:** The heat map highlights customer ordering patterns across weekdays, weekends, and different times of the day, showing that weekend evenings experience the highest demand.
- **Cuisine Preference Analysis:** The highlight table compares cuisine preferences across breakfast, lunch, dinner, and snacks, indicating balanced customer preferences.

Dashboard 2 – Customer Demographics and Spending

- **Age Group Analysis:** The horizontal bar chart shows that customers aged **26–35** contribute the highest number of food orders.
- **Customer Mood Analysis:** The scatter plot compares average order value and customer ratings across different customer moods.

- **Discount Analysis:** The vertical bar chart evaluates the impact of discounts on average order value and shows only a slight difference between discounted and non-discounted orders.

Dashboard 3 – Customer Loyalty and Restaurant Performance

- **Repeat Order Analysis:** The pie chart displays the proportion of repeat and non-repeat orders, showing an almost equal distribution.
- **Restaurant Type Analysis:** The bar chart compares total order values across Budget, Mid-range, and Premium restaurants, with Premium restaurants generating the highest total order value.
- **Weather Impact Analysis:** The line chart compares average order values during rainy and non-rainy weather across different order times, indicating minimal weather impact.

Interactive Dashboard Features

- Interactive filters allow users to analyze data by **City, Age Group, Cuisine, Restaurant Type, Order Time, and Weather Condition**.
- Tooltips provide additional information when users hover over visualizations.
- Dashboards enable quick comparison of customer behavior, ordering trends, and business performance.

Below is the verified public access link to the interactive, cloud-hosted visualization story:

Tableau Public Dashboard Link:

Dashboard 1:

https://public.tableau.com/views/Book1_17850066072090/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Dashboard 2:

https://public.tableau.com/views/Book1_17850066072090/Dashboard2?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

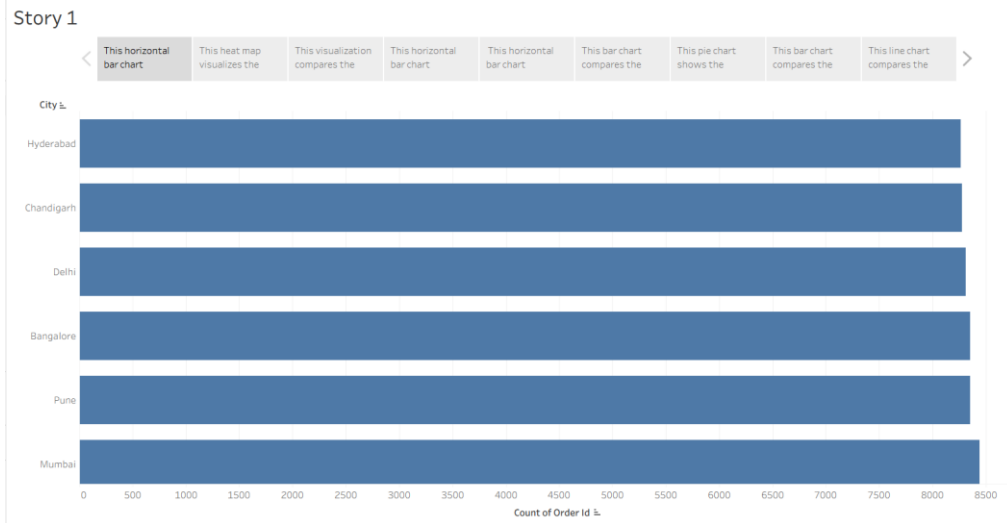
Dashboard 3:

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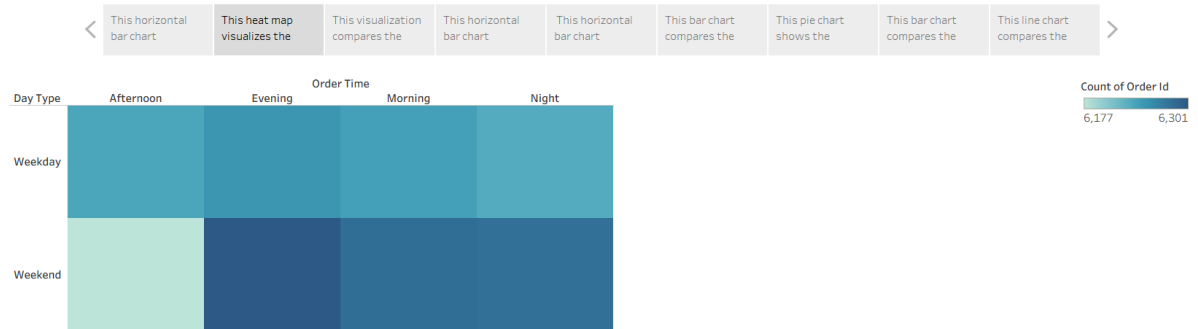
6. Report

6.1. Story Design File

Activity 1: Story Design:



Story 1



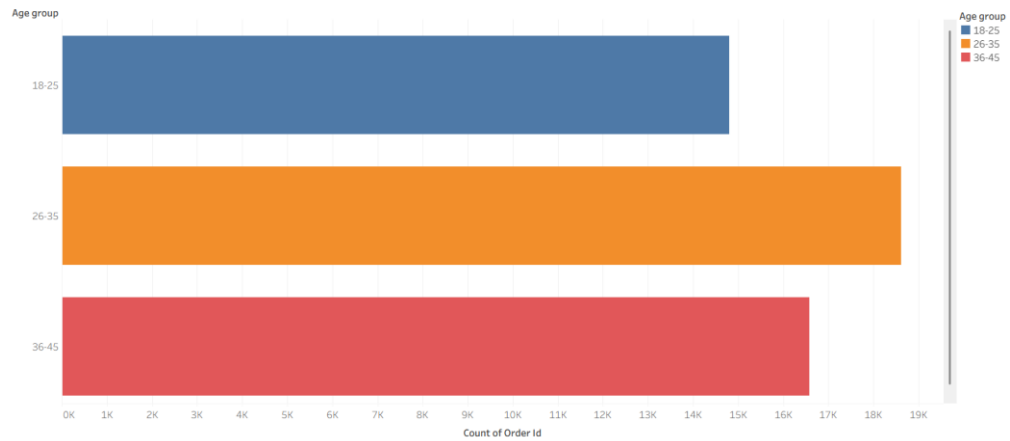
Story 1

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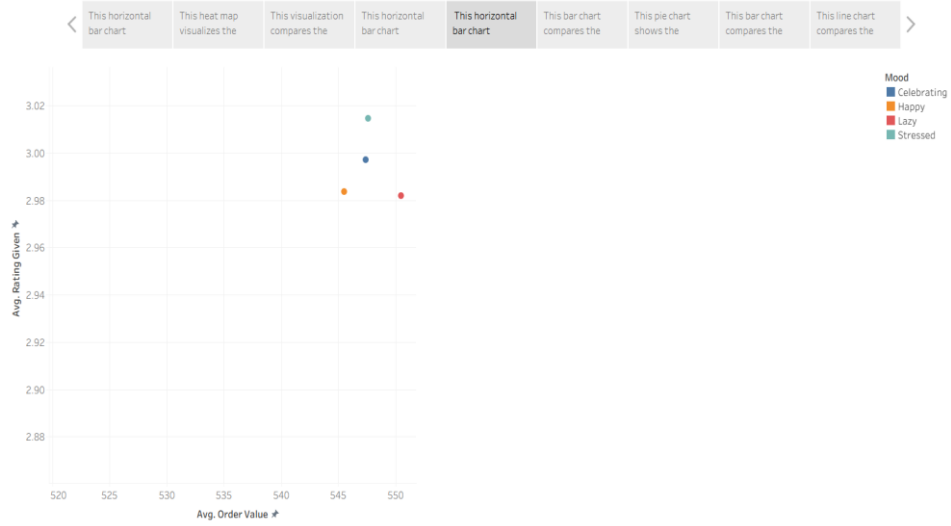


Story 1

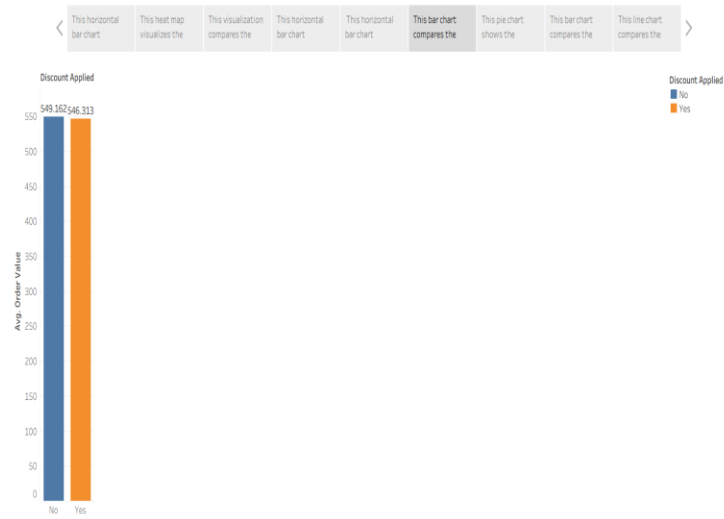
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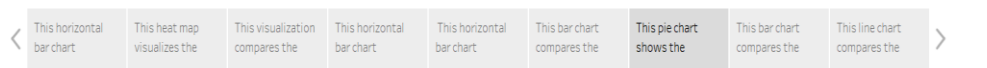
Story 1



Story 1



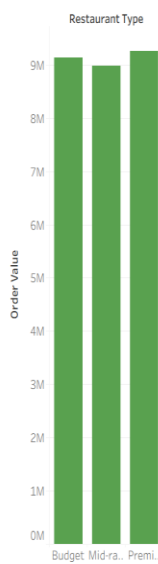
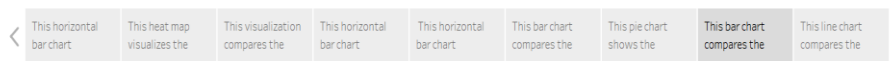
Story 1



Is Repeat Order

- No (Blue)
- Yes (Orange)

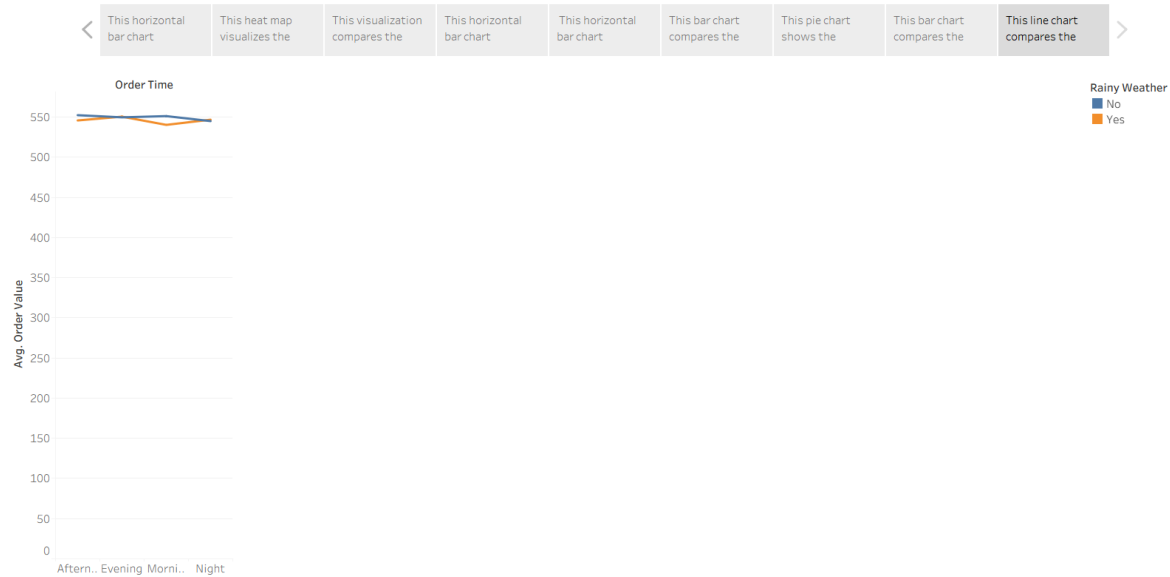
Story 1



Measure Names

- Order Value (Green)

Story 1



Customer Mood and Spending (Task 1): The scatter plot shows only slight variation in average order value and customer ratings across different customer moods.

Customer Ordering Patterns (Task 2): The heat map indicates that weekend evenings and nights experience the highest order volume, making them the busiest ordering periods.

Cuisine Preferences (Task 3): The cuisine comparison visualization shows that customer preferences remain fairly balanced across breakfast, lunch, dinner, and snack meals.

Customer Demographics (Task 4): The age group analysis reveals that customers aged 26–35 contribute the highest number of food orders, making them the primary target customer segment.

City-wise Order Analysis (Task 5): The city comparison chart shows that Mumbai records the highest number of food orders, followed closely by Bangalore and Delhi, helping identify the best-performing markets.

Discount Analysis (Task 6): The comparison of discounted and non-discounted orders indicates that discounts have minimal impact on average order value.

Customer Loyalty (Task 7): The pie chart shows that repeat and non-repeat orders are almost equally distributed, reflecting a balanced mix of returning and first-time customers.

Restaurant Performance (Task 8): The restaurant comparison chart shows that Premium restaurants generate the highest total order value compared to Budget and Mid-range restaurants.

Weather Impact (Task 9): The line chart demonstrates that rainy weather has very little influence on customer order value across different times of the day.

Activity 2:

Publish Story on Tableau Public and Paste Public link below

https://public.tableau.com/views/Book1_17850066072090/Story1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

7. Performance Testing

Performance testing was conducted to ensure that the Tableau dashboard provides a smooth and responsive user experience. The dashboard was evaluated based on the use of filters, calculated fields, and visualizations to ensure efficient data processing and interaction.

7.1 Utilization of Data filters

The dashboard uses multiple interactive filters to help users analyze the data from different perspectives. These filters improve usability by allowing users to narrow down the displayed information without affecting the overall dashboard performance.

Filters used include:

- Restaurant Name
- Cuisine Type
- Order Date
- City/Location

7.2 No of Calculation Field

Several calculated fields were created in Tableau to derive meaningful insights from the raw data. These calculations simplify analysis and support dynamic visualizations.

7.3 No of Visualization

The dashboard contains multiple visualizations designed to present different aspects of the food ordering data. Each visualization addresses a specific business requirement and provides interactive insights.

Visualizations used include:

- Bar Chart
- Line Chart
- Pie/Donut Chart
- Map Visualization
- Tree map
- Heat Map

8. Conclusion/Observation

The Food Ordering Data Visualization Dashboard successfully demonstrates how Tableau can transform raw data into meaningful and interactive visualizations. By using charts, KPIs, calculated fields, and filters, the dashboard helps analyze customer ordering patterns, restaurant performance, delivery efficiency, and overall business trends. It provides valuable insights that support better decision-making and shows how data visualization can simplify complex information for effective business analysis.

9. Future Scope

The dashboard can be improved in the future by integrating real-time data to provide live updates on orders and business performance. Features such as demand forecasting, customer segmentation, and customer review analysis can offer deeper insights. It can also be connected to cloud platforms and mobile devices, allowing users to access reports anytime and anywhere. These enhancements would make the dashboard more effective for supporting business decisions and improving customer satisfaction.

10. Appendix

10.1. Source Code(if any)

This project was developed using **Tableau Desktop**, where dashboards and visualizations were created through Tableau's drag-and-drop interface and calculated fields. No conventional programming source code was used. The project files include:

- Tableau Workbook (.twb/.twbx)
- Dataset (CSV/Excel)
- Calculated Fields and Dashboard Configurations

10.2. GitHub & Project Demo Link

Demo Link :

<https://drive.google.com/file/d/1uKJtH8Yt6lti2gHjrXLZdV60dngqrK8K/view?usp=sharing>

GitHub Link:

<https://github.com/purvij621-dot/Food-ordering-behaviour-and-consumer-trends-structured-analysis-of-choices-and-habits->